Министерство науки и высшего образования Российской Федерации
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ (НИ ТГУ)
Институт прикладной математики и информатики

ОТЧЕТ ПО ИНДИВИДУАЛЬНОМУ ДОМАШНЕМУ ЗАДАНИЮ
по дисциплине «Системы управления базами данных»

Кинотеатр

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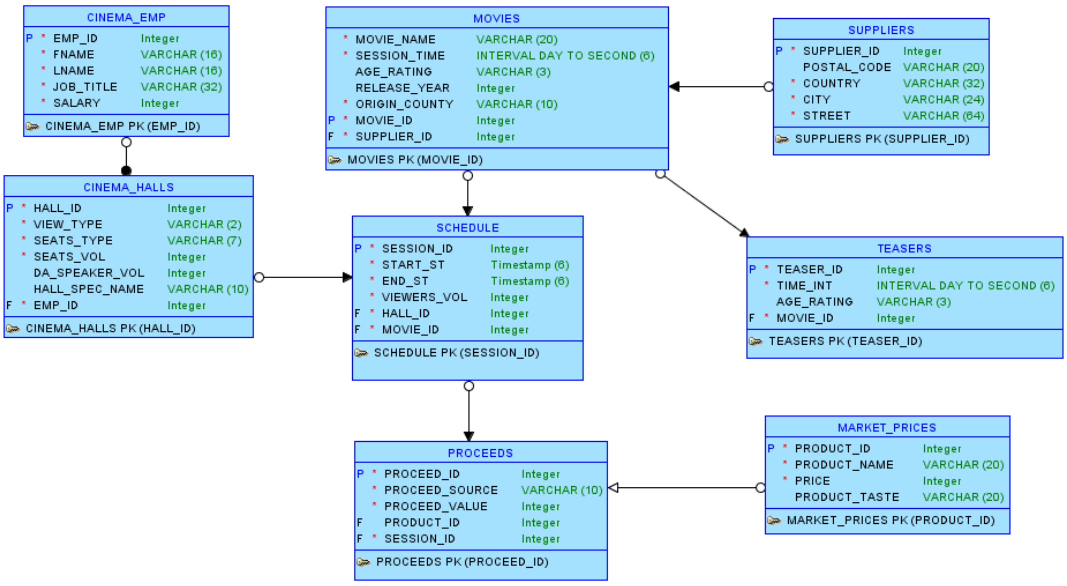
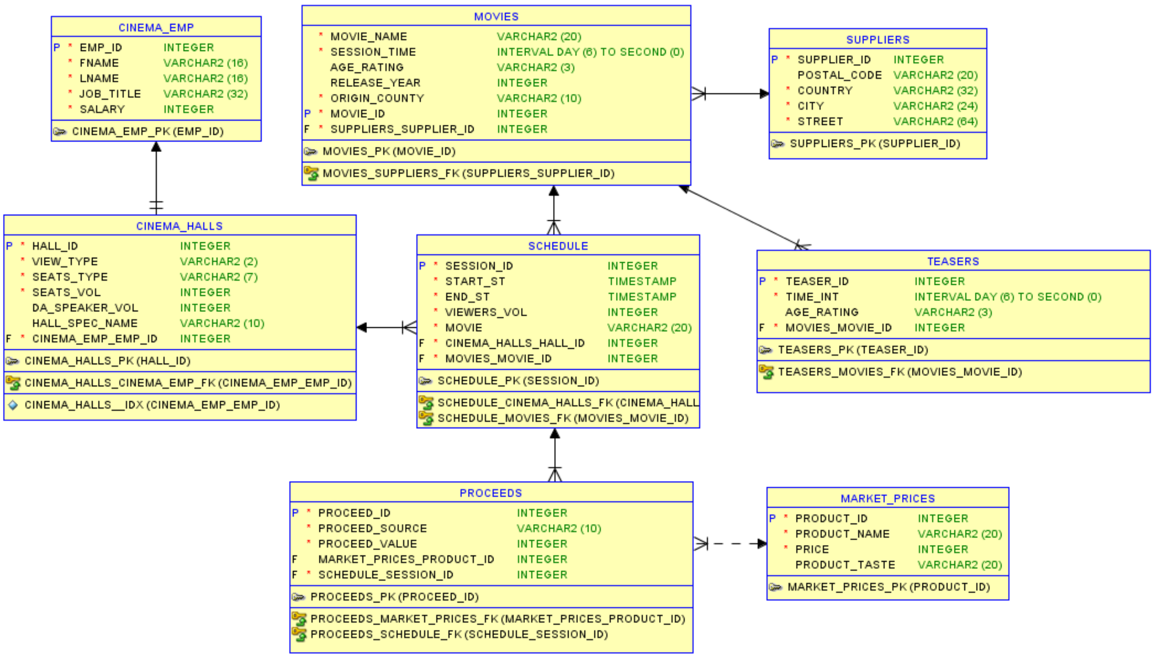
Содержание:

1. Логическая модель.



2. Реляционная модель.



3. DDL.

CREATE TABLE HR.CINEMA_EMP
(EMP_ID NUMBER(3,0),
FNAME VARCHAR2(16 BYTE) CONSTRAINT FNAME_NN NOT NULL,
LNAME VARCHAR2(16 BYTE) CONSTRAINT LNAME_NN NOT NULL,
JOB_TITLE VARCHAR2(32 BYTE),
SALARY NUMBER(5,0),
CONSTRAINT EMP_ID_PK PRIMARY KEY (EMP_ID));

CREATE TABLE HR.CINEMA_HALLS

```sql
  (HALL_ID NUMBER(1,0) CONSTRAINT CINEMA_HALLS_HALL_ID_NN NOT NULL,
   VIEW_TYPE VARCHAR2(2 BYTE) CONSTRAINT CINEMA_HALLS_VIEW_TYPE_NN NOT NULL,
   SEATS_TYPE VARCHAR2(7 BYTE) CONSTRAINT CINEMA_HALLS_SEATS_TYPE_NN NOT NULL,
   FS_EMP_ID NUMBER(3,0) CONSTRAINT CINEMA_HALLS_FS_EMP_ID_NN NOT NULL,
   SEATS_VOL NUMBER(3,0) CONSTRAINT CINEMA_HALLS_SEATS_VOL_NN NOT NULL,
   DA_SPEAKER_VOL NUMBER(2,0),
   HALL_SPEC_NAME VARCHAR2(10 BYTE),
   CONSTRAINT CINEMA_HALLS_SEATS_VOL_CC CHECK ((SEATS_VOL >= 0) and (SEATS_VOL < 100)),
   CONSTRAINT CINEMA_HALLS_HALL_ID_PK PRIMARY KEY (HALL_ID),
   CONSTRAINT "CINEMA_HALLS_UK1" UNIQUE (FS_EMP_ID));

CREATE TABLE HR.MARKET_PRICES
  (PRODUCT_ID NUMBER(2,0),
   PRODUCT_NAME VARCHAR2(20 BYTE) CONSTRAINT MP_PRODUCT_NAME_NN NOT NULL,
   PRICE NUMBER(3,0) CONSTRAINT MP_PRICE_NN NOT NULL,
   PRODUCT_TASTE VARCHAR2(20 BYTE) DEFAULT 'ORIGIN',
   CONSTRAINT MP_PRODUCT_ID_PK PRIMARY KEY (PRODUCT_ID));

CREATE TABLE HR.MOVIES
  (MOVIE_NAME VARCHAR2(20 BYTE) NOT NULL,
   SESSION_TIME INTERVAL DAY (0) TO SECOND (6) CONSTRAINT MOVIES_SESSION_TIME_NN NOT
NULL,
   AGE_RATING VARCHAR2(3 BYTE) DEFAULT '0+',
   SUPPLIER_ID NUMBER(6,0) CONSTRAINT MOVIES_SUPPLIER_NN NOT NULL,
   RELEASE_YEAR NUMBER(4,0),
   ORIGIN_COUNTRY VARCHAR2(10 BYTE) CONSTRAINT MOVIES_ORIGIN_COUNTRY_NN NOT NULL,
   MOVIE_ID NUMBER(3,0),
   CONSTRAINT MOVIES_MOVIE_ID_PK PRIMARY KEY (MOVIE_ID));

CREATE TABLE HR.PROCEEDS
  (PROCEED_ID NUMBER(3,0),
   SESSION_ID NUMBER(3,0) CONSTRAINT PROC_SESSION_ID_NN NOT NULL,
   PROCEED_SOURCE VARCHAR2(10 BYTE) CONSTRAINT PROC_PROCEED_SOURCE_NN NOT NULL,
   PRODUCT_ID NUMBER(3,0),
   PROCEED_VALUE NUMBER(6,0) NOT NULL,
   CONSTRAINT PROC_PROCEED_ID_PK PRIMARY KEY (PROCEED_ID),
   CONSTRAINT PROC_PROCEED_VALUE_NN CHECK (PROCEED_VALUE IS NOT NULL),
   CONSTRAINT PROC_SESSION_ID_FK FOREIGN KEY (SESSION_ID)
   REFERENCES HR.SCHEDULE(SESSION_ID),
   CONSTRAINT PROCEEDS_FK1 FOREIGN KEY (PRODUCT_ID)
   REFERENCES HR.MARKET_PRICES (PRODUCT_ID));
```

```sql
CREATE TABLE HR.SCHEDULE
  (HALL_ID NUMBER(1,0) CONSTRAINT SCHEDULE_HALL_ID_NN NOT NULL,
   START_ST TIMESTAMP (6) CONSTRAINT SCHEDULE_START_ST_NN NOT NULL,
   END_ST TIMESTAMP (6) CONSTRAINT SCHEDULE_END_ST_NN NOT NULL,
   VIEWERS_VOL NUMBER(3,0) CONSTRAINT SCHEDULE_VIEWERS_VO_NN NOT NULL,
   MOVIE_ID NUMBER(3,0),
   SESSION_ID NUMBER(3,0) NOT NULL,
   CONSTRAINT SCHEDULE_VIEWERS_VOL_CC CHECK ((VIEWERS_VOL >= 0) and (VIEWERS_VOL < 100)),
   CONSTRAINT SCHEDULE_UK1 UNIQUE (SESSION_ID),
   CONSTRAINT SCHEDULE_PK PRIMARY KEY (HALL_ID, START_ST),
   CONSTRAINT SCHEDULE_HALL_ID_FK FOREIGN KEY (HALL_ID)
   REFERENCES HR.CINEMA_HALLS (HALL_ID),
   CONSTRAINT SCHEDULE_FK1 FOREIGN KEY (MOVIE_ID)
   REFERENCES HR.MOVIES (MOVIE_ID));

CREATE TABLE HR.SUPPLIERS
  (SUPPLIER_ID NUMBER(6,0),
   POSTAL_CODE VARCHAR2(20 BYTE),
   COUNTRY VARCHAR2(32 BYTE) CONSTRAINT COUNTRY_NN NOT NULL,
   CITY VARCHAR2(24 BYTE) CONSTRAINT CITY_NN NOT NULL,
   STREET VARCHAR2(64 BYTE) CONSTRAINT STREER_NN NOT NULL,
   CONSTRAINT SUPPLIER_ID_PK PRIMARY KEY (SUPPLIER_ID));

CREATE TABLE HR.TEASERS
  (TEASER_ID NUMBER(3,0),
   MOVIE_ID NUMBER(3,0) CONSTRAINT MOVIE_ID_NN NOT NULL,
   TIME_INT INTERVAL DAY (2) TO SECOND (6) CONSTRAINT TIME_INT_NN NOT NULL,
   AGE_RATING VARCHAR2(3 BYTE) DEFAULT '0+',
   CONSTRAINT TEASER_ID_PK PRIMARY KEY (TEASER_ID),
   CONSTRAINT MOVIE_ID_FK FOREIGN KEY (MOVIE_ID)
   REFERENCES HR.MOVIES (MOVIE_ID));
```

4. DML.

```sql
INSERT INTO HR.CINEMA_EMP(FNAME, LNAME, JOB_TITLE, SALARY)
SELECT e.first_name, e.last_name, j.job_title, e.salary
FROM HR.employees e LEFT JOIN HR.jobs j USING(job_id);

INSERT INTO HR.CINEMA_HALLS
VALUES(1,'2D','DEFAULT',70,60,NULL,NULL);
INSERT INTO HR.CINEMA_HALLS
VALUES(2,'3D','ACTION',71,64,8,NULL);
INSERT INTO HR.CINEMA_HALLS
VALUES(3,'2D','DEFAULT',72,90,NULL, PRESS);
INSERT INTO HR.CINEMA_HALLS
VALUES(4,'3D','ACTION',73,80,6,NULL);
INSERT INTO HR.CINEMA_HALLS
VALUES(5,'3D','ACTION',74,90,8,ACTION);
INSERT INTO HR.CINEMA_HALLS
```

```sql
VALUES(6,'2D','DEFAULT',75,98,NULL,NULL);
INSERT INTO HR.CINEMA_HALLS
VALUES(7,'3D','DEFAULT',76,86,4,NULL);
INSERT INTO HR.CINEMA_HALLS
VALUES(8,'3D','DEFAULT',77,80,NULL,NULL);
INSERT INTO HR.CINEMA_HALLS
VALUES(9,'2D','DEFAULT',78,60,2,NULL);

INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'POP CORN', 50, 'ORIGIN');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'POP CORN', 60, 'SALT');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'POP CORN', 65, 'CHEESE');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'CHIPS', 100, 'CHEESE');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'CHIPS', 100, 'BEACON');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'CHIPS', 100, 'SALT');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'CHIPS', 100, 'CRAB');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'COLA CO', 80, 'ORIGIN');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'COLA CO', 90, 'VANILA');
INSERT INTO HR.MARKET_PRICES
VALUES (NULL, 'COLA CO', 80, 'PEPSI');

INSERT INTO HR.MOVIES
VALUES('Knifes out 2',INTERVAL '02:15:00' HOUR TO SECOND,'0+',10,NULL,'UK',4);
INSERT INTO HR.MOVIES
VALUES('Knifes out',INTERVAL '02:35:00' HOUR TO SECOND,'0+',10,NULL,'UK',5);
INSERT INTO HR.MOVIES
VALUES('Fast and furious X',INTERVAL '01:55:00' HOUR TO SECOND,'0+',10,NULL,'USA',6);
INSERT INTO HR.MOVIES
VALUES('Iron gate',INTERVAL '01:23:00' HOUR TO SECOND,'16+',7,2023,'JA',8);
INSERT INTO HR.MOVIES
VALUES('Shadow step 2',INTERVAL '02:45:00' HOUR TO SECOND,'18+',19,2024,'USA',9);


INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
```

```
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'TICKET',NULL, 350);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'CHIPS',4, 100);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'CHIPS',6, 100);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'COLA CO',9, 90);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'COLA CO',9, 90);
INSERT INTO HR.PROCEEDS
VALUES(NULL,4,'COLA CO',10, 80);
INSERT INTO HR.PROCEEDS
VALUES(NULL,1,'TICKET',NULL, 250);
INSERT INTO HR.PROCEEDS
VALUES(NULL,1,'TICKET',NULL, 250);
INSERT INTO HR.PROCEEDS
VALUES(NULL,1,'TICKET',NULL, 250);
INSERT INTO HR.PROCEEDS
VALUES(NULL,1,'TICKET',NULL, 250);
INSERT INTO HR.PROCEEDS
VALUES(NULL,1,'TICKET',NULL, 250);
INSERT INTO HR.PROCEEDS
```

```
VALUES(NULL,1,'COLA CO',9, 90);
INSERT INTO HR.PROCEEDS
VALUES(NULL,1,'POP CORN',2, 60);
INSERT INTO HR.PROCEEDS
VALUES(NULL,1,'POP CORN',3, 65);

INSERT INTO HR.SCHEDULE
VALUES(7,TO_TIMESTAMP('2023/01/16 05:30:00', 'YYYY/MM/DD
HH/MI/SS'),TO_TIMESTAMP('2023/01/16 08:05:00', 'YYYY/MM/DD HH/MI/SS'),5,5,NULL);
INSERT INTO HR.SCHEDULE
VALUES(5,TO_TIMESTAMP('2023/01/16 06:30:00', 'YYYY/MM/DD
HH/MI/SS'),TO_TIMESTAMP('2023/01/16 07:53:00', 'YYYY/MM/DD HH/MI/SS'),0,8,NULL);
INSERT INTO HR.SCHEDULE
VALUES(7,TO_TIMESTAMP('2023/01/16 08:20:00', 'YYYY/MM/DD
HH/MI/SS'),TO_TIMESTAMP('2023/01/16 11:05:00', 'YYYY/MM/DD HH/MI/SS'),20,9,NULL);
INSERT INTO HR.SCHEDULE
VALUES(8,TO_TIMESTAMP('2023/01/16 08:20:00', 'YYYY/MM/DD
HH/MI/SS'),TO_TIMESTAMP('2023/01/16 11:05:00', 'YYYY/MM/DD HH/MI/SS'),80,9,NULL);
INSERT INTO HR.SCHEDULE
VALUES(3,TO_TIMESTAMP('2023/01/16 07:25:00', 'YYYY/MM/DD
HH/MI/SS'),TO_TIMESTAMP('2023/01/16 10:00:00', 'YYYY/MM/DD HH/MI/SS'),80,5,NULL);

INSERT INTO HR.SUPPLIERS (POSTAL_CODE, COUNTRY, CITY, STREET)
SELECT L.postal_code, C.country_name, L.city, L.street_address
FROM HR.locations L LEFT JOIN HR.countries C USING(country_id);

INSERT INTO HR.TEASERS
VALUES(NULL,4,INTERVAL '00:01:43' HOUR TO SECOND, '0+');
INSERT INTO HR.TEASERS
VALUES(NULL,4,INTERVAL '00:02:10' HOUR TO SECOND, '0+');
INSERT INTO HR.TEASERS
VALUES(NULL,8,INTERVAL '00:02:13' HOUR TO SECOND, '16+');
INSERT INTO HR.TEASERS
VALUES(NULL,9,INTERVAL '00:01:42' HOUR TO SECOND, '18+');
INSERT INTO HR.TEASERS
VALUES(NULL,5,INTERVAL '00:01:41' HOUR TO SECOND, '0+');
INSERT INTO HR.TEASERS
VALUES(NULL,5,INTERVAL '00:02:07' HOUR TO SECOND, '0+');
INSERT INTO HR.TEASERS
VALUES(NULL,5,INTERVAL '00:01:31' HOUR TO SECOND, '0+');
```

5. Индексы

На все первичные ключи и уникальные столбцы Oracle автоматически создает уникальные индексы.

```
CREATE TABLESPACE MY_INDEXES DATAFILE 'MY_INDEXES.dat'
SIZE 50M REUSE AUTOEXTEND ON NEXT 50M MAXSIZE 1000M;
```

```
CREATE UNIQUE INDEX MOVIES_MOVIE_NAME_IDX ON HR.MOVIES(MOVIE_NAME)
TABLESPACE MY_INDEXES;
CREATE INDEX SUP_ID_IDX ON HR.MOVIES(SUPPLIER_ID)
TABLESPACE MY_INDEXES;

CREATE INDEX PROC_SESSION_ID_IDX ON HR.PROCEEDS(SESSION_ID)
TABLESPACE MY_INDEXES;

CREATE UNIQUE INDEX SCHEDULE_SESSION_IDX ON HR.SCHEDULE(HALL_ID, START_ST, END_ST)
TABLESPACE MY_INDEXES;
CREATE INDEX MOVIE_ID_IDX ON HR.SCHEDULE(MOVIE_ID)
TABLESPACE MY_INDEXES;

CREATE INDEX TEAS_MOVIE_ID_IDX ON HR.TEASERS(MOVIE_ID)
TABLESPACE MY_INDEXES;
```

6. Последовательности и триггеры.

```
CREATE SEQUENCE CINEMA_EMP_ID_SQ START WITH 1;
CREATE OR REPLACE TRIGGER CINEMA_EMP_ID_TGG BEFORE INSERT OR UPDATE ON HR.CINEMA_EMP
FOR EACH ROW
BEGIN
IF INSERTING AND :NEW.EMP_ID IS NULL THEN :NEW.EMP_ID := CINEMA_EMP_ID_SQ.NEXTVAL;
end if;
end;

CREATE SEQUENCE CINEMA_HALLS_ID_SQ START WITH 1;
CREATE OR REPLACE TRIGGER CINEMA_HALLS_ID_TGG BEFORE INSERT OR UPDATE ON
HR.CINEMA_HALLS
FOR EACH ROW
BEGIN
IF INSERTING AND :NEW.HALL_ID IS NULL THEN :NEW.HALL_ID := CINEMA_HALLS_ID_SQ.NEXTVAL;
end if;
end;

CREATE SEQUENCE MARKET_PRICES_ID_SQ START WITH 1;
CREATE OR REPLACE TRIGGER MARKET_PRICES_ID_TGG BEFORE INSERT OR UPDATE ON
HR.MARKET_PRICES
FOR EACH ROW
BEGIN
```

```
IF INSERTING AND :NEW.PRODUCT_ID IS NULL THEN :NEW.PRODUCT_ID :=
MARKET_PRICES_ID_SQ.NEXTVAL;
end if;
end;

CREATE SEQUENCE MOVIE_ID_SEQUENCE START WITH 1;
CREATE OR REPLACE TRIGGER MOVIE_ID_TRIG BEFORE INSERT OR UPDATE ON HR.MOVIES
FOR EACH ROW
BEGIN
IF INSERTING AND :NEW.MOVIE_ID IS NULL THEN :NEW.MOVIE_ID := MOVIE_ID_SEQUENCE.NEXTVAL;
end if;
end;

CREATE SEQUENCE PROCEEDS_ID_SQ START WITH 1;
CREATE OR REPLACE TRIGGER PROCEEDS_ID_TGG BEFORE INSERT OR UPDATE ON HR.PROCEEDS
FOR EACH ROW
BEGIN
IF INSERTING AND :NEW.PROCEED_ID IS NULL THEN :NEW.PROCEED_ID := PROCEEDS_ID_SQ.NEXTVAL;
end if;
end;

CREATE SEQUENCE SESSION_ID_SQ START WITH 1;
CREATE OR REPLACE TRIGGER SESSION_ID_TGG BEFORE INSERT OR UPDATE ON HR.SCHEDULE
FOR EACH ROW
BEGIN
IF INSERTING AND :NEW.SESSION_ID IS NULL THEN :NEW.SESSION_ID := SESSION_ID_SQ.NEXTVAL;
end if;
end;

CREATE SEQUENCE SUPPLIERS_ID_SQ START WITH 1;
CREATE OR REPLACE TRIGGER SUPPLIERS_ID_TGG BEFORE INSERT OR UPDATE ON HR.SUPPLIERS
FOR EACH ROW
BEGIN
IF INSERTING AND :NEW.SUPPLIER_ID IS NULL THEN :NEW.SUPPLIER_ID := SUPPLIERS_ID_SQ.NEXTVAL;
end if;
end;

CREATE SEQUENCE TEASERS_ID_SQ START WITH 1;
CREATE OR REPLACE TRIGGER TEASERS_ID_TGG BEFORE INSERT OR UPDATE ON HR.TEASERS
FOR EACH ROW
BEGIN
IF INSERTING AND :NEW.TEASER_ID IS NULL THEN :NEW.TEASER_ID := TEASERS_ID_SQ.NEXTVAL;
```

```
end if;
end;
```

7. Представления.

Представление, в котором находится необходимая информация для противопожарной безопасности:

```
CREATE OR REPLACE VIEW HR.FS_INFO AS SELECT s.hall_id, c.fs_emp_id, s.viewers_vol, s.start_st, s.end_st
FROM HR.SCHEDULE S LEFT JOIN HR.CINEMA_HALLS C ON S.HALL_ID = C.HALL_ID
LEFT JOIN HR.MOVIES M ON S.MOVIE_ID = M.MOVIE_ID
WHERE s.start_st < LOCALTIMESTAMP AND s.end_st > LOCALTIMESTAMP;
```

Представление, в котором находится информация о всей выручке с каждого сеанса:

```
CREATE VIEW HR.PROSEED_INFO AS SELECT S.session_id, S.hall_id, S.movie, H.view_type,
(SELECT SUM(PROCEED_VALUE) FROM HR.PROCEEDS WHERE SESSION_ID = S.SESSION_ID) AS SUM
FROM HR.Schedule S LEFT JOIN HR.Cinema_halls H ON S.hall_id = H.hall_id;
```